

Tom Derrick

Data Scientist — Electricity Markets & Quantitative Analytics · Sydney, Australia · tderrick98@outlook.com ·
github.com/tder311 · linkedin.com/in/tderrick98

PROFILE

Data scientist with four years in Australia's National Electricity Market, across price forecasting, mathematical optimisation, and applied AI. I've worked the market from both sides — inside a generator's trading and risk function at Delta Electricity, and now leading the modelling behind Modo Energy's Australian forecast products. My work is production analytics that decisions get made on: MILP dispatch and capacity models solving over the real transmission network, ML price forecasters, contract valuation engines, and retrieval-grounded LLM systems built with evaluation and audit trails rather than demos. I direct the NEM model roadmap, reporting to the Head of Region, and train and delegate to a junior data scientist.

EXPERIENCE

Data Scientist — Modo Energy, Sydney

Aug 2025 – present

Series B scaleup building forecasting and investment intelligence for the energy transition — customers include Goldman Sachs, Apollo, EDF, and Shell. I joined the Australian team in its first year and lead the modelling behind its local products.

- Designed and shipped the nodal transmission network model behind the July 2026 NEM forecast release — DC optimal power flow, bus-level dispatch, PTDF, load disaggregation, transmission augmentation through 2050+. Took the product from regional to 200+ nodes, so assets are valued at the connection point where they actually sit: largest update since the Australian launch and its first on-time release, delivered as sole modeller while training a new data scientist.
- Designed the capacity expansion model (linopy + XPRESS MILP) determining least-cost generation and storage buildout across the NEM, replacing weeks of manual capacity tuning per forecast with a single optimisation.
- Build and run the production cost models producing long-term NEM price forecasts; battery dispatch optimisation and revenue modelling in the multi-market Global Dispatch Model (Pyomo + XPRESS).
- Built the contracting strategy system end to end — a valuation engine pricing PPAs, tolls and offtake structures against forecast dispatch, plus the React configuration UX shipped to customers.
- Built and deployed an LLM support agent (Claude + RAG over model documentation, code, and model outputs) answering customer and internal questions across Slack and Intercom — retrieval across heterogeneous sources, with answers grounded in retrieved context rather than free generation.
- Contributed data engineering and semantic table annotation to Modo's customer-facing AI analyst, so the model interprets forecast datasets correctly.
- Act as the forecasting expert on customer calls alongside the sales team — technical due diligence with prospects evaluating the product, and discovery with existing customers whose requirements feed back into the NEM model roadmap I direct.
- Present each forecast release at Modo's customer workshops and to the wider business — translating model and methodology changes into what they mean for investment and trading decisions.
- Author Modo's published forecast research — model releases, market outlooks, and deep dives.

Market Analyst — Delta Electricity, Sydney

Jul 2022 – Jul 2025

NSW electricity generator — front-office analytics supporting trading, hedging, and portfolio risk.

- Designed whole-of-market NEM simulation models (PLEXOS/Prophet) used for trading, hedging, and business development decisions.

- Built a Monte-Carlo Gross-Margin-at-Risk engine quantifying portfolio earnings risk across price and volume scenarios, used for risk compliance and portfolio optimisation.
- Built an ML-powered spot-price forecasting model supporting short-term trading decisions.
- Developed LGC and ACCU certificate trading strategy across environmental commodity markets.
- Modelled major investment decisions, including the end-of-life extension of Vales Point Power Station.
- Optimised the planned-outage schedule to minimise revenue impact; built the internal live trading-performance dashboard. Completed Delta's 2024 future-leadership program.

Customer Success Manager — TechTarget, Sydney

Nov 2020 – Jun 2022

- Managed APAC campaign delivery for marquee accounts (Red Hat, Dell, IBM, Cisco, Alibaba).

PERSONAL PROJECTS

- **nem-market-watch** (nemdash.up.railway.app) — live dashboard for Australia's National Electricity Market: five-minute prices and demand, market metrics, PASA reserve outlook, bid-band analysis, and a tool-calling LLM agent that answers questions against live data. React · FastAPI · PostgreSQL · Railway.
- **nem-forward** (forward-frontend-production.up.railway.app) — XGBoost price forecaster (7-day, half-hourly, P10–P90) benchmarked against AEMO pre-dispatch, LP battery dispatch optimiser, and bid-band generation; every forecast persisted with its model's training timestamp so past runs are scored against actuals instead of overwritten. XGBoost · FastAPI · React.
- **nem-ingestion** — five-minute AEMO market data pipeline: six NEMWEB clients, sole writer and schema owner across 13 tables, backfill CLIs, hourly rollups, and a retention sweep that cut the database from 21 GB to 2 GB. Python · PostgreSQL · Railway.
- **agent-tui** (github.com/tder311/agent-tui) — terminal UI for local AI coding agents: git worktrees, branches, Claude Code / OpenCode sessions, live running agents, and open PRs, grouped by project via app-agnostic filesystem discovery; read-only by default. Go · Bubble Tea.
- **opencode-orchestrator** (github.com/tder311/opencode-orchestrator · npm) — opencode plugin for parallel AI coding agents: each spawns into an isolated git worktree with its own interactive terminal session, pushing completion/blocker reports back to a supervising session.
- **railway-tui** (github.com/tder311/railway-tui) — terminal dashboard for Railway in Go and Bubble Tea: single static binary, OAuth (PKCE) login, live view of deployments, log streams, and environment variables over Railway's GraphQL API, with restart/redeploy/rollback actions and rate-limit-aware polling.

EDUCATION & ACCREDITATION

- **AFMA Financial Markets Accreditation**
- UNSW — Master of Data Science (with Excellence)
- University of Sydney — BSc, Computer Science & Statistics

SKILLS

Languages: Python · SQL · Go · JavaScript

Technical: NEM market modelling · PLEXOS · linopy · Pyomo · XPRESS (MILP/LP) · XGBoost · time-series forecasting · Monte-Carlo simulation & risk modelling · RAG & LLM systems (Claude, OpenAI) · AWS (Batch, S3) · FastAPI · React · PostgreSQL · git

Soft: Client-facing technical due diligence · presenting to technical and non-technical audiences · mentoring and delegating to junior analysts · roadmap ownership and prioritisation · published technical writing